PCRQ 12

Graded

Student

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Total Points

3 / 5 pts

Question 1

(no title) **0** / 1 pt

+ 1 pt Correct

✔ **+ 0 pts** Incorrect

Question 2

(no title) **0** / 1 pt

+ 1 pt Correct

✔ **+ 0 pts** Incorrect

Question 3

(no title) **1** / 1 pt

✔ **+ 1 pt** Correct

+ 0 pts Incorrect

Question 4

(no title) **1** / 1 pt

✔ **+ 1 pt** Correct

+ 0 pts Incorrect

Question 5

(no title) **1** / 1 pt

✔ **+ 1 pt** good-faith answer

+ 0 pts blank

looking at figure 1 might help to see this- we need to find a ball around z that is contained inside the ball B(c,r)-- so this radius makes this work.

Q1
1 Point

The interval $[1, 2) \subseteq \mathbb{R}$ (with the usual metric $d(x, y) = |x - y|$) is:

- ○ open
- ○ closed
- ◉ both open and closed
- ○ neither open nor closed

Q2
1 Point

True or false: The set $\{3\}$ is not an open subset of $\mathbb{R}$, no matter which metric we put on $\mathbb{R}$.

- ◉ True
- ○ False

Q3
1 Point

Each interval $\left(-\frac{1}{n}, \frac{1}{n}\right)$ is open in $\mathbb{R}$. The intersection $\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right)$ is:

- ○ open, since an intersection of open sets is always open
- ◉ $\{0\}$, which is not open
- ○ $\varnothing$, since the intervals shrink to nothing
- ○ $\{0\}$, which is open

Q4
1 Point

A subset of $\mathbb{R}$ that is not open, is closed.

- ○ True
- ◉ False

Q5

1 Point

Please write briefly (in two or three sentences) what single point of the notes of Lecture 13 you found to be the most difficult or confusing. If you did not find any part of it difficult or confusing, please say what part(s) you found most interesting.

I have question regarding Proposition 2.3. I'm confused of why do we choose to define ε := r−d(z, c) particularly and from where is it based on?